

Agentic Reasoning for Large Language Models: A Survey

Team Name
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Chapter 1

Introduction

The emergence of large language models (LLMs) capable of autonomous, multi-step reasoning has opened a new research frontier: *agentic systems* that perceive their environment, plan sequences of actions, and interact with external tools to accomplish complex goals [1]. Unlike single-turn question answering, agentic LLMs must maintain context across many steps, recover from errors, and coordinate with other agents or subsystems [2].

1.1 Motivation

Classical approaches to agent design relied on hand-crafted planners and symbolic state representations. Modern LLM-based agents replace the symbolic backbone with a pre-trained model whose internal representations encode broad world knowledge [1]. This shift enables rapid adaptation to novel tasks but introduces new challenges around reliability, memory, and grounding.

1.2 Scope and Contributions

This survey covers four core capabilities of reasoning-centric agentic LLMs: planning and task decomposition, memory and state persistence, retrieval-augmented grounding, tool use, and multimodal reasoning. We also discuss evaluation frameworks and open research directions.

1.3 Overview in Hebrew /

This section demonstrates correct bidirectional text rendering under LuaLaTeX with polyglossia and the David CLM Hebrew font.

מערכות LLM אגנטיות מאפשרות ביצוע משימות מורכבות על ידי תכנון, זיכרון ושימוש בכלים חיצוניים. הסקר הזה סוקר את הגישות המובילות בתחום: planning,

multimodal rea-) וחשוב רב-מודאלי (tool use), שימוש בכלים (retrieval, memory soning). מחקרים כמו [1] מניחים את היסודות לפיתוח סוכנים חכמים יותר.

The paragraph above is typeset right-to-left, with embedded English technical terms switching back to left-to-right rendering automatically via polyglossia.

1.4 Chapter Outline

Chapter 2 surveys planning and task decomposition. Chapter 3 covers memory architectures. Chapter 4 discusses retrieval-augmented reasoning. Chapter 5 examines tool-use policies. Chapter 6 addresses multimodal agentic systems. Chapter 7 reviews evaluation benchmarks. Chapter 8 summarises findings and future directions.

Chapter 2

Planning and Task Decomposition

Planning is the capacity of an agent to decompose a high-level goal into a sequence of achievable sub-tasks and to select actions that advance toward that goal [3]. Frontier LLMs have shown remarkable improvements in planning by leveraging chain-of-thought reasoning and search-based exploration.

2.1 Formal Objective

Let $\mathcal{A}$ denote the set of agents operating in a state space $\mathcal{S}$ with action space $\mathcal{U}$. A policy $\pi : \mathcal{S} \rightarrow \mathcal{U}$ induces a trajectory $\tau = (s_0, a_0, s_1, a_1, \dots, s_T)$. The planning objective is to find the policy that maximises cumulative reward:

$$\pi^* = \arg \max_{\pi} \mathbb{E} \left[\sum_{t=0}^T \gamma^t R(s_t, a_t) \right] \quad (2.1)$$

where $\gamma \in (0, 1]$ is a discount factor and R is the scalar reward received at each step. As expressed in Equation (2.1), the agent must balance immediate and future rewards — a challenge that hierarchical decomposition addresses by reducing the effective horizon.

Definition 2.1 (Hierarchical Task Decomposition). *A hierarchical task decomposition of goal g is a tree of sub-goals $\{g_1, \dots, g_k\}$ such that, for any state in which all g_i have been achieved, the agent has also achieved g .*

2.2 Benchmark Evidence

[3] evaluated frontier LLMs on classical planning benchmarks (Blocksworld, ALFWorld, GSM8K) and found that search-augmented models outperform

pure chain-of-thought by a significant margin. Table 2.1 summarises the key results.

Table 2.1: Comparison of planning approaches in frontier LLM agents.

Method	Strategy	Benchmark	Score
GPT-4o	Chain-of-thought	Blocksworld	74%
o1-preview	Search + backtrack	Blocksworld	81%
Claude 3.5	ReAct	ALFWorld	67%
Gemini 1.5 Pro	Tree-of-thought	GSM8K	95%

Scores from [3]. Higher is better; benchmarks are not directly comparable across rows.

2.3 The Agentic Perception–Action Loop

Figure 2.1 illustrates the core agentic loop in which an agent repeatedly observes its environment, reasons about the next action, and executes that action.

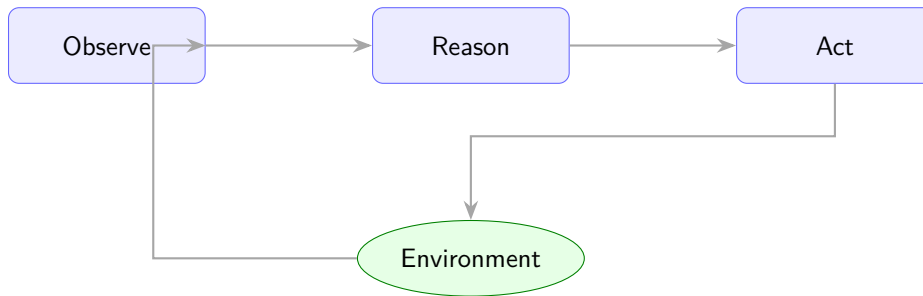


Figure 2.1: The agentic perception–action loop. Arrows show the cyclic flow: the agent *observes* state, *reasons* to select an action, and *acts* on the environment, which transitions to a new state.

2.4 Graph: Planning Benchmark Comparison

Figure 2.2 shows the planning benchmark comparison generated by the Phase 8 graph pipeline (P8-I01).

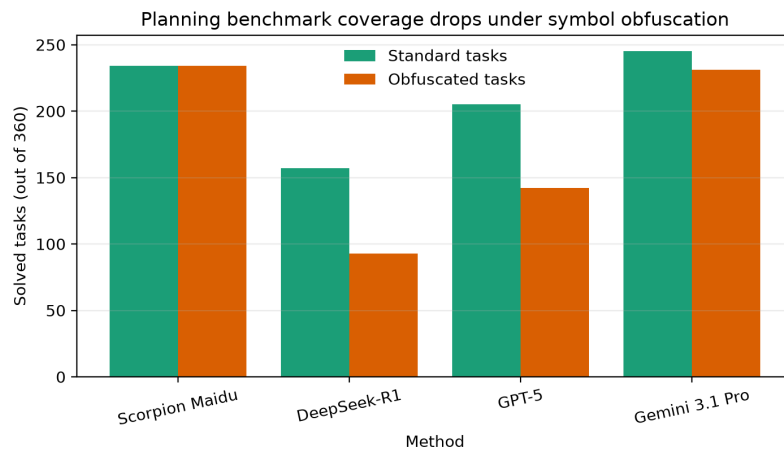


Figure 2.2: Planning benchmark comparison across frontier LLMs. Generated deterministically by the Phase 8 visualisation pipeline.

Chapter 3

Memory and State Persistence

Long-horizon agentic tasks require agents to retain information across many steps and interaction turns. Modern LLM-based agents address this by maintaining explicit memory stores alongside the model’s implicit parametric knowledge [4, 5].

3.1 Memory Taxonomies

Agent memory is commonly classified along two axes:

- **Episodic memory** stores raw experience traces (observations, actions, rewards) and supports retrieval by similarity or recency.
- **Semantic memory** stores distilled facts and concepts extracted from experience, enabling efficient generalisation.
- **Working memory** holds the current context window of the LLM and is the primary bottleneck for long-horizon tasks.

3.2 Lightweight Cognitive Architectures

LiCoMemory [4] introduces a lightweight cognitive memory system that reduces storage overhead while preserving recall accuracy on long-reasoning benchmarks. TeleMem [5] extends this to multimodal long-term memory, maintaining both textual and visual traces across extended interaction histories.

3.3 Memory Mechanisms in Hebrew /

This section demonstrates Hebrew/English bidirectional typesetting. The body text is Hebrew (right-to-left); embedded English technical terms switch direction inline via `polyglossia's \textenglish` command.

מנגנוני זיכרון הם מרכיב מרכזי בכל מערכת LLM אגנטית. הסוכן (agent) חייב לשמור מידע לאורך פרקי זמן ארוכים כדי לבצע משימות מורכבות. גישות כמו LiCoMemory [4] ו-TeleMem [5] מציגות ארכיטקטורות זיכרון קלות משקל שמגדילות את יכולת הסוכן לשמור מידע ולאחזר אותו בצורה יעילה. שיטות אלו מאזנות בין עלות האחסון (storage cost) לבין דיוק האחזור (recall accuracy).

The paragraph above is typeset in Hebrew with correct right-to-left alignment and bidirectional text flow. English citations and terms appear left-to-right as inline switches, satisfying the PRD §8.6 BiDi requirement.

Chapter 4

Retrieval-Augmented Reasoning

Retrieval-augmented generation (RAG) grounds LLM outputs in external knowledge sources, reducing hallucination and enabling the agent to reason over information beyond its training cut-off [6].

4.1 Reactive vs. Proactive Retrieval

Early RAG systems retrieved documents only in response to an explicit query at inference time. Recent work on proactive retrieval [7] anticipates information needs before the LLM formulates a query, pre-fetching relevant context and reducing latency on multi-step tasks.

4.2 Retrieval in Medical Multimodal Settings

[7] demonstrated that proactive retrieval improves diagnostic accuracy when applied to multimodal medical LLMs that must reason over both clinical text and radiological images. The key insight is that retrieval decisions can be made by a lightweight routing model that monitors agent state, rather than waiting for the base LLM to request information explicitly.

4.3 Agentic Retrieval Frameworks

[6] frame retrieval as an agentic sub-policy: the agent learns when to retrieve, what to retrieve, and how to integrate retrieved content into its current reasoning chain. This formulation connects retrieval to the policy objective in Equation (2.1) of Chapter 2: the agent maximises expected reward by selecting retrieval actions only when the information gain justifies the latency cost.

Chapter 5

Tool Use and External Interfaces

Tool-augmented LLMs extend the basic perception–action loop (Figure 2.1) with the ability to invoke external interfaces: calculators, code interpreters, search engines, and domain-specific APIs [6].

5.1 Bounded Tool-Use Policies

A bounded tool-use policy constrains the agent to a finite, pre-declared set of tools $\mathcal{T}$, each with a typed schema. The agent selects a tool $t \in \mathcal{T}$ and supplies structured arguments; the tool returns an observation that is appended to the agent’s context window. This keeps the action space tractable and makes the agent’s behaviour auditable [6].

5.2 Mathematical Reasoning with Tools

[8] showed that LLMs equipped with a Python interpreter as a tool dramatically outperform pure chain-of-thought on competition-level mathematics. The agent generates code, executes it, observes the result, and revises its solution — a verify-and-revise loop that mirrors the iterative problem-solving strategy of human mathematicians.

5.3 Tool Use and the Planning Objective

Tool invocations can be modelled as a special class of action in the planning objective of Equation (2.1): a tool call $a_t \in \mathcal{T}$ transitions the state by adding new information to the agent’s context, yielding a high immediate reward when the retrieved information resolves an ambiguity or enables a correct next step.

Chapter 6

Multimodal Agentic Reasoning

Agentic systems are increasingly required to reason over heterogeneous modalities — text, images, audio, and structured data — within a single interaction [9]. Multimodal LLMs unify these modalities in a shared representation space, enabling cross-modal evidence fusion.

6.1 Agentic Multimodal Architectures

[9] survey the design space of agentic multimodal LLMs along four axes: perception (how modalities are encoded), grounding (how encodings are aligned to language), reasoning (how evidence from multiple modalities is fused), and action (how the agent produces outputs or invokes tools based on multimodal context).

The key architectural choice is whether modalities are fused early (at the embedding layer), late (at the output layer), or dynamically (the model selects a fusion strategy per step). Dynamic fusion consistently outperforms static alternatives on tasks that require selective attention to a single modality.

6.2 Proactive Multimodal Retrieval

As discussed in Chapter 4, proactive retrieval strategies are especially effective in multimodal settings, where the cost of retrieving an image is substantially higher than retrieving a text snippet [7]. The routing model must therefore weight the expected information gain against the retrieval cost when deciding whether to fetch a visual asset.

6.3 Open Challenges

Key open problems include: (i) compositional reasoning across more than two modalities simultaneously; (ii) long-context multimodal memory (connecting Chapter 3 to this chapter); and (iii) evaluation protocols that test multimodal reasoning independently of language understanding.

Chapter 7

Evaluation Frameworks

Rigorous evaluation of agentic LLMs is challenging because agent performance depends not only on final-answer accuracy but also on intermediate reasoning quality, tool-use efficiency, and robustness to distribution shift [10].

7.1 The MIRAI Benchmark

MIRAI [10] is a benchmark for evaluating LLM agents on event forecasting: the agent must predict future geopolitical events by reasoning over a dynamic knowledge graph constructed from news articles. Unlike static QA benchmarks, MIRAI requires the agent to retrieve time-sensitive information, chain multi-hop inferences, and express calibrated uncertainty — capabilities that directly correspond to the skills surveyed in Chapters 4–6.

7.2 Dimensions of Agent Evaluation

A comprehensive agent evaluation framework should measure at least four dimensions:

1. **Task completion rate** — fraction of episodes in which the agent achieves the goal within the allotted steps.
2. **Efficiency** — number of steps, tokens, and tool calls consumed per episode.
3. **Robustness** — performance under observation noise, misleading tool outputs, or adversarial prompts.
4. **Calibration** — correlation between the agent’s expressed confidence and its empirical accuracy.

7.3 Connections to the Planning Objective

From the planning perspective of Equation (2.1), evaluation reduces to measuring how closely the agent’s achieved cumulative reward approximates π^* . Current benchmarks typically proxy reward with a binary task-success indicator, which conflates good planning with lucky execution. Future evaluation designs should adopt denser reward signals that credit correct intermediate reasoning steps.

Chapter 8

Conclusion

This survey has examined five core capabilities of reasoning-centric agentic LLMs: planning, memory, retrieval, tool use, and multimodal reasoning. Across all five, a consistent theme emerged: agents that decompose tasks into bounded sub-problems, maintain explicit state, and ground decisions in retrieved evidence substantially outperform agents that rely on a single, monolithic forward pass [1, 2].

8.1 Key Findings

1. **Planning:** Search-augmented frontier LLMs approach or match state-of-the-art classical planners on standard benchmarks [3], suggesting that the gap between neural and symbolic planning is narrowing rapidly.
2. **Memory:** Lightweight cognitive architectures ([4, 5]) provide an effective middle ground between context-window memory and external databases, enabling long-horizon tasks without prohibitive storage overhead.
3. **Retrieval:** Proactive retrieval strategies reduce latency and improve grounding accuracy, particularly in high-stakes domains [7].
4. **Tool use:** Bounded tool-use policies combined with code-execution loops enable LLMs to solve tasks that require exact computation [8].
5. **Multimodal:** Dynamic modality-fusion architectures consistently outperform static alternatives, pointing toward unified agentic reasoners that treat all input channels symmetrically [9].

8.2 Future Directions

Open problems span all five capability axes. For planning, learning to set task horizons adaptively remains unsolved. For memory, efficient forgetting

— retaining task-relevant facts while discarding irrelevant detail — is an active research question. For retrieval, the alignment between retrieval relevance and downstream task reward deserves more theoretical treatment. For tool use, compositional policies that chain multiple tools in a single reasoning step are in their infancy. For multimodal reasoning, benchmarks that isolate cross-modal inference from language understanding are needed.

Progress on each of these fronts will bring agentic LLMs closer to the vision articulated by [1]: systems that reason, plan, and act with the generality and robustness required for open-ended, real-world deployment.

Nomenclature

$\mathcal{A}$	Set of agents in a multi-agent system
π	Policy mapping observations to action distributions
$\mathcal{S}$	State space of the planning environment
V	Value function estimating cumulative expected reward

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